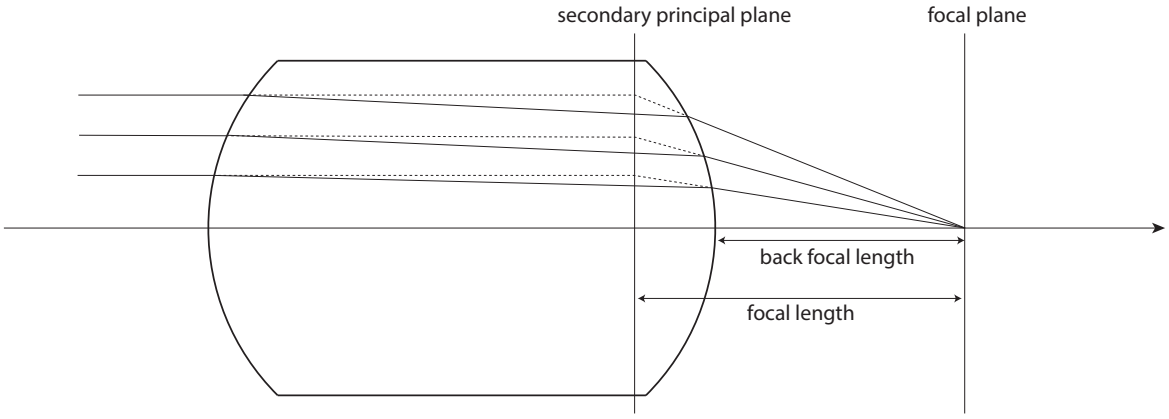
Quantum Optics – Problem Set No. 1

Ray optics and ABCD matrices

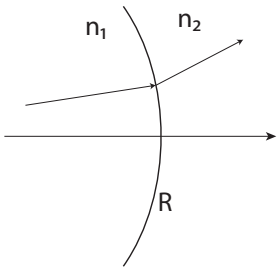
(1.1) Thick lens

In the lecture, we introduced the ABCD matrix for a thin lens. Here, your goal is to derive the ABCD matrix for a thick lens, including its focal length (effective focal length) and back focal length.

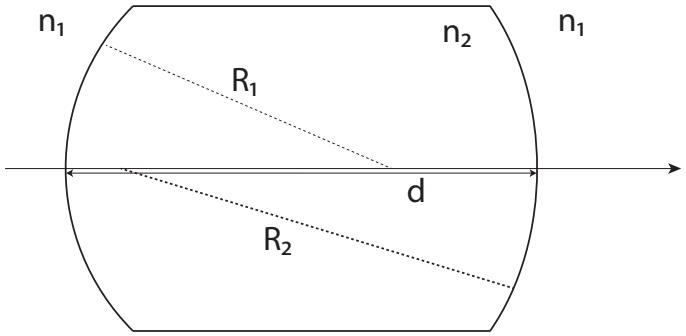
(a) First, let get ourselves familiar with the concept of the *back focal length* and the *focal length* (sometimes also called effective focal length). Consider an optical element described by an ABCD matrix that focuses parallel incident rays to one point (in the focal plane), as shown below. The focal position measured from the back surface of the optical element is called the *back focal length*. Find the back focal length in dependence on the relevant ABCD matrix parameters. The effect of focusing can be represented by placing a thin lens at the secondary principal plane. Derive the expression for the focal length measured from the secondary principal plane in terms of the relevant ABCD matrix parameters.



(b) Consider a spherical interface with a radius of curvature of R as shown below. Using Snell's law at the interface, derive the ABCD matrix for the transmitted rays under the paraxial approximation.



(c) Combining the ABCD matrices for spherical interfaces (result from (b)) and free space propagation (following the convention used in our lecture, the ABCD matrix for propagation in a medium of constant refractive index is the same as in free space), derive the ABCD matrix for a thick lens shown below.



(d) Combining the results from (a) and (c), give the expressions for the back focal length and the focal length of a thick lens. Show that for a plano-convex lens ($R_2 = \infty$), the difference between the focal length and the back focal length is given by $d \times n_1/n_2$. Show that the ABCD matrix for a thick lens converges to that of a thin lens when the distance between the interfaces d is small compared to the radii R_1 and R_2.

(e) Calculate the back focal length and the focal length of the following three lenses placed in free space ($n_1 = 1$) for the design wavelength of $\lambda = 588\,\text{nm}$ ($n_2 = 1.4584$ for fused silica glass, $n_2 = 1.5168$ for N-BK7 glass, which you can check on the webpage refractiveindex.info):

- fused silica plano-convex lens with $R_1 = 4.1\,\text{mm}$, $R_2 = \infty$, $d = 1.5\,\text{mm}$
- fused silica bi-convex lens with $R_1 = R_2 = 8.8\,\text{mm}$, $d = 2.6\,\text{mm}$
- N-BK7 "best form" lens with $R_1 = 24.0\,\text{mm}$, $R_2 = 134.6\,\text{mm}$, $d = 6.5\,\text{mm}$

(The above lenses are Thorlabs part no. LA4039, LB4280, LBF254-040, respectively. You can check whether your results agree with the specifications by typing "Thorlabs" followed by the part no. into Google, going to the Thorlabs webpage of the part number and downloading the "Auto CAD PDF" file which gives the focal length and the back focal length.)

(f) A transmissive spherical object works as a lens (ball lens). Derive the focal length and back focal length of a ball lens using the results obtained above.

(g) A retroreflector is a device that reflects all the incoming rays into the direction of its source. What is the condition that a ball lens works as a retroreflector when the back-side hemisphere is reflective?

(1.2) Resonator stability: modified confocal design due to manufacturing errors

Imagine that you designed a stable confocal cavity with a pair of mirrors with nominal radii of 200 mm. After you received the ordered mirrors, you found that your cavity is unstable. It turned out, that due to manufacturing errors, these mirrors have radii of 205 mm and 195 mm. How could you modify your cavity design to build a stable cavity from these two mirrors?

Gaussian beam

(1.3) Complex beam parameter representation of a Gaussian beam

In the lecture, we derived the expression for the electric field of a Gaussian beam:

$$E(r,z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{r^2}{w(z)^2} - \mathrm{i}\frac{kr^2}{2R(z)} + \mathrm{i}\arctan(z/z_0) - \mathrm{i}kz\right), \tag{1}$$

where w_0 is the radius of the waist located at $z = 0$, $w(z) = w_0\sqrt{1+(z/z_0)^2}$ is the beam radius, $R(z) = z(1+(z_0/z)^2)$ is the wavefront curvature, and $z_0 = kw_0^2/2$ (where k is the wave number) is the Rayleigh length (related to the confocal parameter $b = 2z_0$). Show that the above expression is identical to the following compact expression:

$$E(r,z) = E_0 \frac{q_0}{q(z)} \exp\left(-\mathrm{i}\frac{kr^2}{2q(z)} - \mathrm{i}kz\right), \tag{2}$$

where $q(z) = z + \mathrm{i}\, z_0 \equiv z + q_0$ is the complex beam parameter. You may want to remind and convince yourself of the following useful expression for the inverse of $q(z)$:

$$\frac{1}{q(z)} = \frac{1}{R(z)} - \mathrm{i}\frac{2}{kw(z)^2}. \tag{3}$$

(1.4) Knife-edge method for beam radius measurement

One way to measure the beam radius $w(z)$ of a Gaussian beam is to advance a knife edge into the beam with a micrometer and measure the power of the beam with a power meter downstream from the knife edge. Between what two percentages of the power will the knife edge advance a distance equal to $w(z)$?

Hint: You can use a computer algebra software (e.g. the online webpage www.wolframalpha.com) to evaluate the integrals, or perform an analytical calculation by expressing the result in terms of the error function:

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \mathrm{d}t, \tag{4}$$

which has the following exemplary values: $\mathrm{erf}(\infty) = 1$, $\mathrm{erf}(1/\sqrt{2}) \simeq 0.6827$.